

ITA0202- WEB TECHNOLOGY
UNIT IV - REPRESENTING WEB DATA
ASSESSMENT I: DATA INTERPRETATION

Question 1: Interpret the XML Structure

- a. Identify the root element.
- b. Identify the repeating record element.
- c. Identify the attribute used to uniquely identify each course.
- d. Identify the elements that represent numeric information.
- e. State whether the XML document is structurally well-formed and justify your answer.

ANSWERS

- a. Root element: <courses>
- b. Repeating record element: <course>
- c. Unique identifier attribute: id — the attribute on <course> (e.g., id="C101", id="C102" ...), used to uniquely identify each course record.
- d. Elements representing numeric information: <students> and <credits>
- e. Is the document well-formed?

Yes.

Justification:

- It has exactly one root element (<courses>)
- Every tag that opens has a matching closing tag (e.g., <course>...</course>)
- Tags are properly nested — no element closes before an element it contains
- The attribute value is quoted (id="C101")
- There are no illegal characters and no unclosed elements

Question 2: Apply XPath for Data Selection

Write suitable XPath expressions to extract each of the following:

- a. All course records.
- b. Names of all courses.
- c. Courses having more than 50 students.

- d. Courses carrying 4 credits.
- e. Courses whose type is Theory.
- f. Names of Theory courses having more than 50 students.
- g. Faculty members handling courses with at least 4 credits.
- h. The course whose id is C104.
- i. The first course available in the XML document.
- j. The last course available in the XML document.

ANSWERS

#	Requirement	XPath Expression
a	All course records	/courses/course
b	Names of all courses	/courses/course/name
c	Courses with more than 50 students	/courses/course[students > 50]
d	Courses carrying 4 credits	/courses/course[credits = 4]
e	Courses of type Theory	/courses/course[type = 'Theory']
f	Names of Theory courses with >50 students	/courses/course[type='Theory' and students>50]/name
g	Faculty of courses with at least 4 credits	/courses/course[credits >= 4]/faculty
h	The course with id C104	/courses/course[@id='C104']
i	First course in the document	/courses/course[1]
j	Last course in the document	/courses/course[last()]

Verified Results

- a. Returns all 5 <course> nodes (C101–C105)
- b. Web Technology, Artificial Intelligence, Web Technology Laboratory, Machine Learning, Database Systems
- c. WEB301 (58), AI302 (72), ML304 (64)
- d. WEB301, AI302, ML304
- e. WEB301, AI302, ML304, DB305
- f. Web Technology, Artificial Intelligence, Machine Learning
- g. Dr. Arun, Dr. Meena, Dr. Priya
- h. ML304 – Machine Learning (Dr. Priya, 64 students)
- i. WEB301 – Web Technology (C101)
- j. DB305 – Database Systems (C105)

Question 3: Apply XSLT for Data Presentation

Develop an XSLT stylesheet that transforms the XML data into an HTML table with the columns Course

Code, Course Name, Faculty, Students, Credits, and Type.

- a. Display only courses having more than 40 students.
- b. Sort the displayed courses in descending order of student enrollment.
- c. Display a suitable heading such as “High Enrollment Courses”.
- d. Use an XPath condition inside the XSLT transformation.
- e. Produce valid HTML output.

ANSWERS

XSLT Stylesheet:

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="1.0"
  xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

<xsl:output method="html" indent="yes" encoding="UTF-8"/>

<xsl:template match="/courses">
<html>
<head>
  <title>High Enrollment Courses</title>
</head>
<body>
  <h2>High Enrollment Courses</h2>
  <table border="1" cellpadding="5" cellspacing="0">
    <tr>
      <th>Course Code</th>
      <th>Course Name</th>
      <th>Faculty</th>
      <th>Students</th>
      <th>Credits</th>
      <th>Type</th>
    </tr>
    <!-- XPath condition: courses with more than 40 students -->
    <xsl:for-each select="course[students > 40]">
      <!-- Sort descending by student enrollment -->
      <xsl:sort select="students" order="descending" data-type="number"/>
      <tr>
        <td><xsl:value-of select="code"/></td>
        <td><xsl:value-of select="name"/></td>
        <td><xsl:value-of select="faculty"/></td>
        <td><xsl:value-of select="students"/></td>
        <td><xsl:value-of select="credits"/></td>
        <td><xsl:value-of select="type"/></td>
      </tr>
```

```

    </xsl:for-each>
  </table>
</body>
</html>
</xsl:template>

</xsl:stylesheet>

```

How Each Requirement Is Satisfied

Requirement	Where It Is Implemented
a. Only courses with >40 students	course[students > 40] in the select attribute of <xsl:for-each>
b. Sorted descending by enrollment	<xsl:sort select="students" order="descending" data-type="number"/>
c. Suitable heading	High Enrollment Courses
d. XPath condition used	course[students > 40] is itself an XPath predicate expression
e. Valid HTML output	<xsl:output method="html"/> plus properly structured <html><head>...</head><body>...</body></html> and a well-formed <table>

Resulting HTML Table (Actual Transformation Output)

Course Code	Course Name	Faculty	Students	Credits	Type
AI302	Artificial Intelligence	Dr. Meena	72	4	Theory
ML304	Machine Learning	Dr. Priya	64	4	Theory
WEB301	Web Technology	Dr. Arun	58	4	Theory
DB305	Database Systems	Dr. Kumar	42	3	Theory

Command to run this with a standard XSLT processor:

```
xsltproc transform.xsl courses.xml > output.html
```

Question 4: Interpret the Extracted Data (5 Marks)

- Identify the course with the highest enrollment.
- Identify the course with the lowest enrollment.
- Determine the number of Theory courses.
- Identify all courses having exactly 4 credits.
- If an additional teaching assistant is assigned to every course with more than 60 students, identify the courses that require additional support.

ANSWERS

- Highest enrollment: AI302 – Artificial Intelligence (72 students, Dr. Meena)
- Lowest enrollment: WEB303 – Web Technology Laboratory (36 students, Dr. Ravi)
- Number of Theory courses: 4 — WEB301, AI302, ML304, DB305 (WEB303 is Practical)

d. Courses with exactly 4 credits: WEB301, AI302, ML304 (3 courses)

e. Courses needing an additional TA (students > 60):

- AI302 – Artificial Intelligence (72 students)
- ML304 – Machine Learning (64 students)